Churn Analysis & Retention Strategy

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1. Problem Definition

Objective:
Identify customer churn patterns, quantify churn rate, analyze key drivers, and propose actionable retention strategies.

Churn Definition:
A customer is considered *churned* if they stop using a service within a given time period.

2. Dataset (Sample Used)

We'll use a **synthetic telecom churn dataset** (commonly used in analytics practice).

Features:

- customerID
- gender
- SeniorCitizen
- tenure
- MonthlyCharges
- TotalCharges
- Contract (Month-to-month, One year, Two year)
- PaymentMethod
- InternetService
- Churn (Yes/No)

3. Step-by-Step Methodology (with Theory)

Step 1: Data Loading & Cleaning

Theory:

Raw datasets often contain:

- Missing values
- Incorrect data types
- Outliers

Cleaning ensures reliability of insights.

Key Actions:

- Convert TotalCharges → numeric
- Handle null values
- Encode categorical variables

Step 2: Churn Rate Calculation

Formula:

$$
\text{Churn Rate} = \frac{\text{Number of churned customers}}{\text{Total customers}} \times 100
$$

Insight:

- High churn rate = business problem
- Benchmark telecom churn: ~15–25%

Step 3: Exploratory Data Analysis (EDA)

Theory:

EDA helps uncover:

- Patterns
- Trends
- Relationships

Key Questions:

- Which customers churn more?
- Does tenure affect churn?
- Are pricing and contracts important?

Step 4: Correlation / Feature Importance

Theory:

Helps identify variables that influence churn most.

Top Factors:

- Contract type
- Tenure
- Monthly charges

- Internet service type

Correlation Heatmap (Interpretation)

Strong indicators:

- Contract (Month-to-month) → positive correlation with churn
- Tenure → negative correlation with churn

Step 5: Customer Segmentation

Segments Identified:

High Risk (Churned)

- Month-to-month contract
- High monthly charges
- Low tenure

Medium Risk

- Moderate tenure
- Mixed contract types

Low Risk (Loyal)

- Long tenure
- Long-term contracts

Segmentation Visualization

Segment	Characteristics	Action Needed
High Risk	New + expensive plans	Immediate retention
Medium	Mid-tenure users	Engagement
Low Risk	Loyal customers	Upsell

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6. Key Insights

1. **Contract type is the strongest churn driver**
2. **Customers churn early (low tenure)**
3. **Higher pricing increases churn**

4. **Long-term customers rarely churn**

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7. Retention Strategy Report

Strategy 1: Convert Monthly Users to Annual Plans

- Offer discounts for long-term contracts
- Example: 20% off for yearly plans

Strategy 2: Early Lifecycle Engagement

- First 90 days onboarding
- Personalized emails, support calls

Strategy 3: Pricing Optimization

- Introduce flexible pricing tiers
- Offer bundles

Strategy 4: Loyalty Programs

- Reward long-term users
- Points, cashback, upgrades

Strategy 5: Predictive Alerts

- Flag high-risk customers
- Trigger retention campaigns

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8. Full Python Notebook Code (.ipynb Equivalent)

You can directly paste this into Jupyter.

```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load dataset
df = pd.read_csv('Telco-Customer-Churn.csv')

# Data Cleaning
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df.dropna(inplace=True)

# Convert target variable
df['Churn'] = df['Churn'].map({'Yes':1, 'No':0})

# ------------------------------
# 1. Churn Rate
# ------------------------------
churn_rate = df['Churn'].mean() * 100
print(f"Churn Rate: {churn_rate:.2f}%")

# ------------------------------
# 2. Churn Distribution
# ------------------------------
sns.countplot(x='Churn', data=df)
plt.title("Churn Distribution")
plt.show()

# ------------------------------
# 3. Contract vs Churn
# ------------------------------
sns.countplot(x='Contract', hue='Churn', data=df)
plt.title("Churn by Contract Type")
plt.show()
```

```
# ----------------------------
# 4. Monthly Charges vs Churn
# ----------------------------
sns.boxplot(x='Churn', y='MonthlyCharges', data=df)
plt.title("Monthly Charges vs Churn")
plt.show()

# ----------------------------
# 5. Tenure vs Churn
# ----------------------------
sns.boxplot(x='Churn', y='tenure', data=df)
plt.title("Tenure vs Churn")
plt.show()

# ----------------------------
# 6. Correlation Heatmap
# ----------------------------
numeric_df = df.select_dtypes(include=['int64','float64'])

plt.figure(figsize=(10,6))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()

# ----------------------------
# 7. Customer Segmentation
# ----------------------------
def segment(row):
    if row['tenure'] < 12 and row['MonthlyCharges'] > 70:
        return 'High Risk'
```

```
    elif row['tenure'] < 24:
        return 'Medium Risk'
    else:
        return 'Low Risk'

df['Segment'] = df.apply(segment, axis=1)

sns.countplot(x='Segment', hue='Churn', data=df)
plt.title("Customer Segmentation")
plt.show()

# Save processed dataset
df.to_csv("processed_churn_data.csv", index=False)
```

📦 9. Final Deliverables (What You Submit)

📁 **Files:**

- ✅ Jupyter Notebook (.ipynb)
- ✅ Processed dataset (.csv)
- ✅ Report (PDF or PPT)

📊 **In Report Include:**

1. Problem statement
2. Dataset description
3. Churn rate calculation
4. Visualizations
5. Insights
6. Segmentation
7. Retention strategies

🧭 If You Want Next Level Upgrade

I can extend this into:

- Machine learning churn prediction (Logistic Regression / Random Forest)
- Dashboard (Power BI / Tableau)
- Real-time churn monitoring system

Just tell me.